Artificial Intelligence and Machine Learning

(6CS012)

Part I

Question and Answer Report

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1. Long Question

1.1. Challenges in Deploying Machine Learning Systems in Production

Operationalizing machine learning (ML) systems in production systems brings a range of technical and business challenges that can influence model accuracy, efficiency, and effectiveness. A key challenge is data drift, where the statistical properties of real-world data differ from the training data set. This can be caused by seasonal effects, user behavior or business process changes. Addressing data drift early is crucial to ensure the model's predictions remain reliable. To mitigate this problem, companies have continuous monitoring pipelines that monitor the input data features and model predictions. The model can then be automatically retrained with new data on a regular basis. But this approach incurs more costs in infrastructure and engineering efforts.

Another common challenge is model latency, especially in real-time decision-making use cases like fraud detection, recommendations, or autonomous monitoring. This slows down the system and degrades performance. To address this challenge, system engineers may make use of efficient model structures, batch inference, and acceleration schemes using GPUs for inference or quantizing models. These techniques enhance performance and efficiency, but can sometimes come at the cost of accuracy.

The third issue encountered in production systems is model decay or drift. Models trained on old data might not perform well as the environment changes. This can result in suboptimal business decisions if not tracked correctly. To address this, companies use monitoring dashboards and alerting systems, which monitor model accuracy metrics and trigger retraining when metrics fall below a certain threshold. While monitoring mechanisms enhance system accuracy, they add to the system's complexity and demand inter-team collaboration.

Data scientists, machine learning engineers, and product managers work closely together to successfully deploy ML systems. Data scientists develop models and interpret performance metrics, engineers handle deployment and scalability issues, and product managers ensure the system meets business goals.

Contemporary technologies such as automated MLOps workflows, cloud computing infrastructure, and large language model–driven systems also improve deployment efficiency through continuous integration, monitoring and retraining. These tools assist organizations in building resilient and responsive ML systems that can be used in production for the long term.

2. Short Question

2.1. Techniques to Reduce Overfitting in Deep Learning

Overfitting is a common issue in machine learning and deep learning where the model overfits the training data and does not perform well on new data. This causes the model to learn irrelevant features and patterns in the data. This leads to high accuracy on the training set but low accuracy on the validation set. Overfitting can be alleviated by two popular techniques, dropout and early stopping, which enhance generalization.

Dropout is a regularization method that randomly turns off a percentage of neurons during training. Dropout reduces the model's dependence on any particular feature or connection by randomly deactivating neurons during training. This promotes the development of more robust feature representations rather than overfitting to the training samples. Dropout is particularly useful in deep convolutional neural networks (CNNs) for visual pattern recognition, like object classification and scene understanding. For instance, when a CNN is trained to classify images of natural scenes, dropout helps the model learn spatial features that are relevant instead of memorizing the background.

Another key technique for avoiding overfitting during model training is early stopping. This involves monitoring the validation loss during training. If the model's performance on the validation dataset fails to improve for a number of training epochs, then the training process is terminated before the model begins overfitting the training dataset. This helps the model achieve the best performance while avoiding over-training. Early stopping is commonly applied in text classification, sentiment analysis, and financial forecasting scenarios where the training data can be noisy or have limited labeled samples.

Dropout and early stopping techniques make model training more robust and help to build models that not only work on the training dataset but also real-life unseen data.

2.2. Differences Between Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)

Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are two widely used deep learning models that can be used to process different data structures and solve different machine learning tasks. CNNs are generally applied to spatial data like images, while RNNs are used for sequential data like speech, text, and time-series data.

CNNs perform spatial operations on the input data using convolutional filters that are applied at various locations across an image to extract significant spatial patterns such as edges, textures, and shapes. These allow the model to automatically extract hierarchical features from simple structures to complex semantic structures. CNNs employ parameter sharing and local connectivity, which greatly reduces the number of parameters and computational cost while retaining spatial information. For these reasons, CNNs are extensively used for image classification, including face recognition, medical image analysis and scene classification.

On the other hand, RNNs are particularly suited for processing sequences, as they have hidden states that carry information about what has happened in the past. This enables them to capture the dependencies between words or signals in the sequence. For this reason, they are frequently applied in tasks like sentiment analysis, speech recognition, language translation and stock price prediction.

Training deep neural networks like CNNs and RNNs can be a challenging task due to issues such as the vanishing gradient problem, where gradients have very small values during backpropagation, making it hard for the earlier layers to update their weights effectively. This can be overcome with methods such as batch normalization, gradient clipping, or more specialized network architectures such as Long Short-Term Memory (LSTM) networks. Overfitting is another common problem that can be tackled by regularization methods such as dropout, early stopping and data augmentation.

Both CNNs and RNNs are critical components in today's artificial intelligence systems, and the choice of network architecture is often determined by the nature of the data being processed.